

Landsat derived Land Surface Temperature climatology data for the analysis of *Vitis vinifera* L. vegeto-productive anomalies.

A case study in the Regional Park of Colli Euganei

ENRICO MAGAZZINO,

PhD Candidate, TESAF

SUPERVISORS:

FRANCO MEGGIO, PHD

CARLO ZANETTI

Research aims:

- ✓ Investigating the Relationship between **LST Anomalies** and **Grapevine Productivity Anomalies**
- ✓ Identification of most **Sensible Regions Using Risk Maps**
- ✓ Influence of **Temperature Anomalies** on Different Grape Varieties' **Buds' Fruitfulness**
- ✓ Development of possible Adaptive Management Practices

Land Surface Temperature (LST) from Satellite

LST data provides a powerful tool in modern agriculture for improving resource management, predicting crop performance, and adapting to changing climatic conditions.

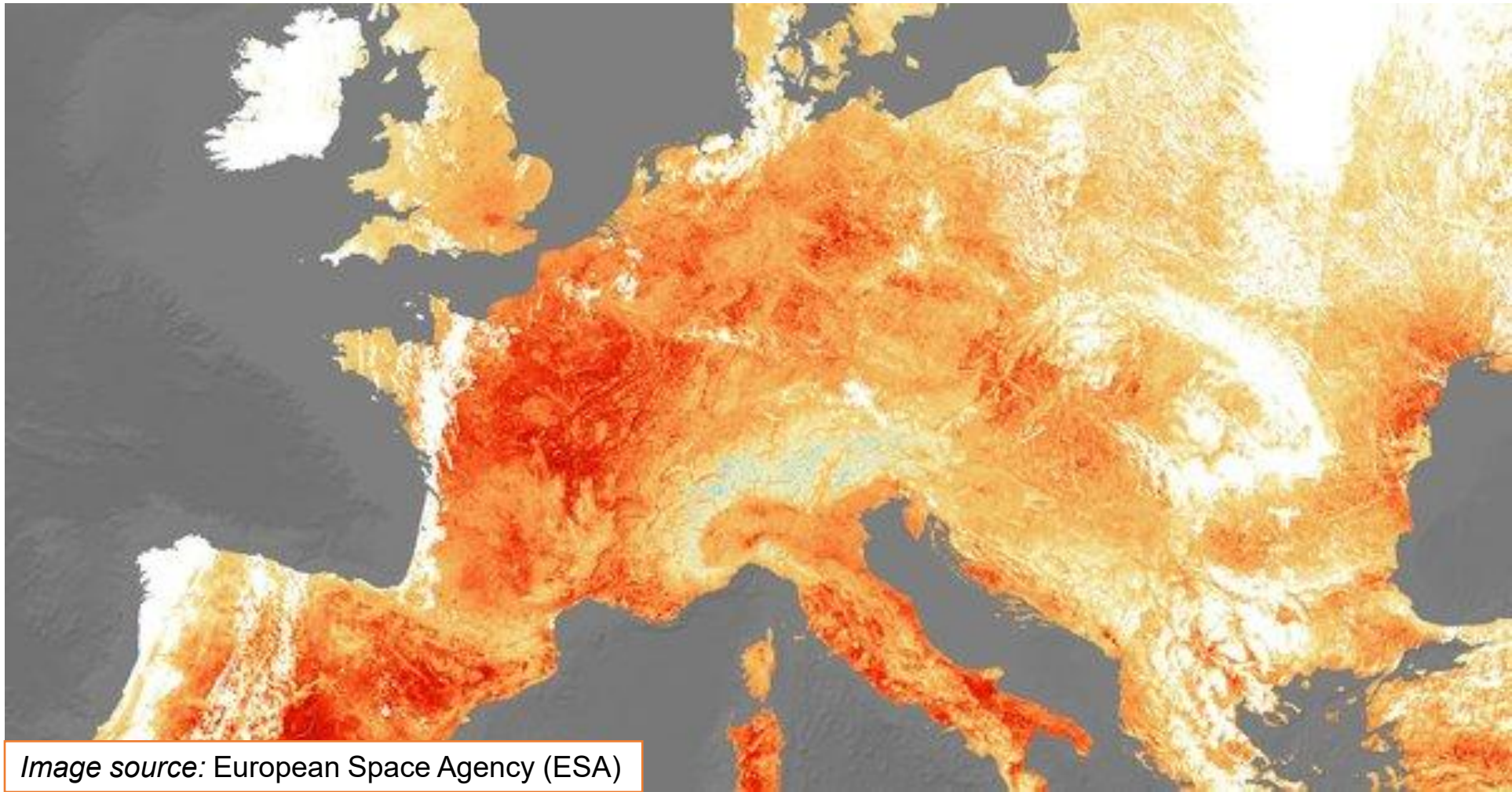


Image source: European Space Agency (ESA)

Applications of LST in Agriculture

- **Crop Health Monitoring**
- **Water Management**
- **Frost and Heat Risk Assessment**
- **Yield Estimation**
- **Climate Change and Adaptation**
- **Pest and Disease Management**

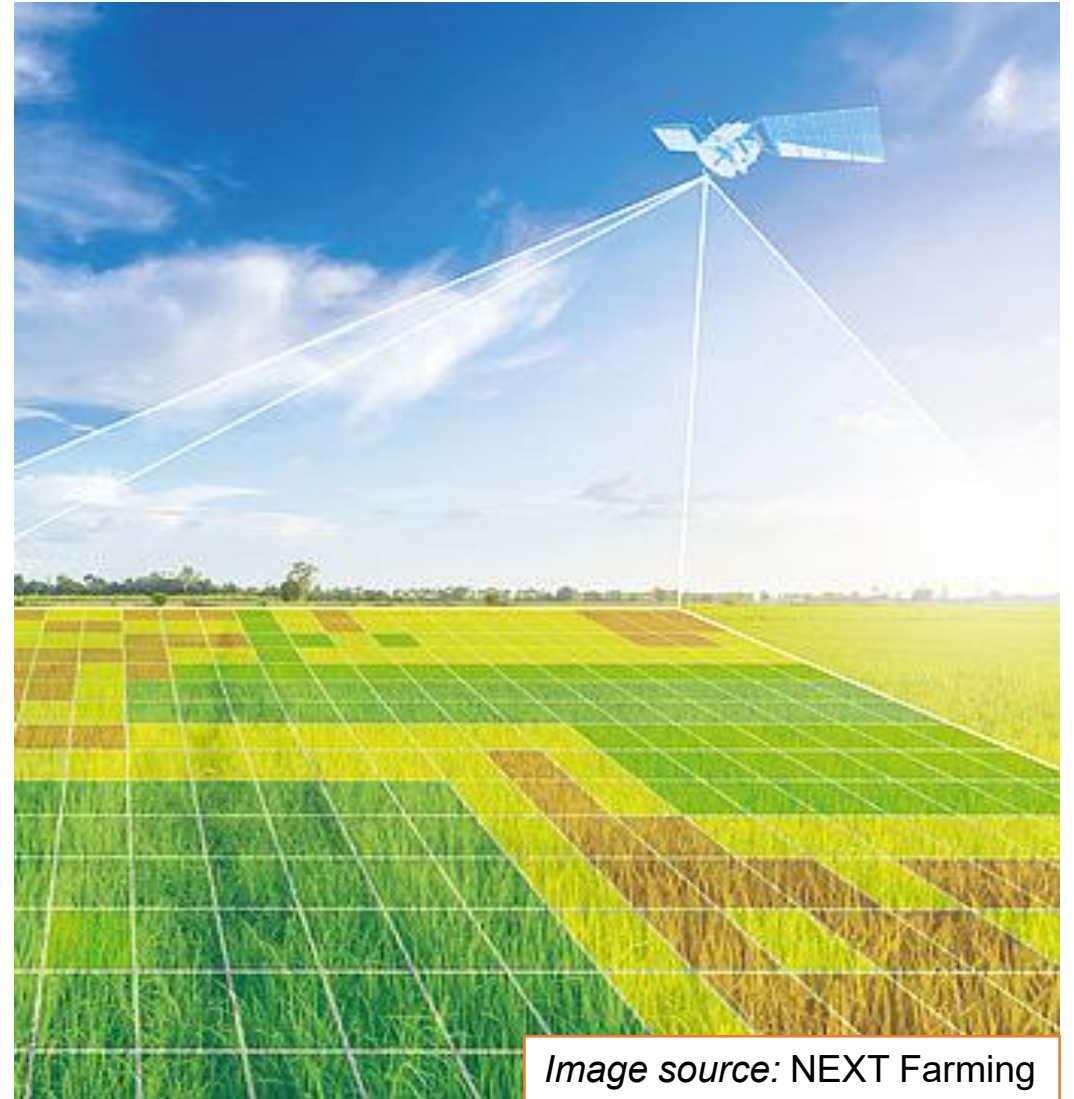


Image source: NEXT Farming

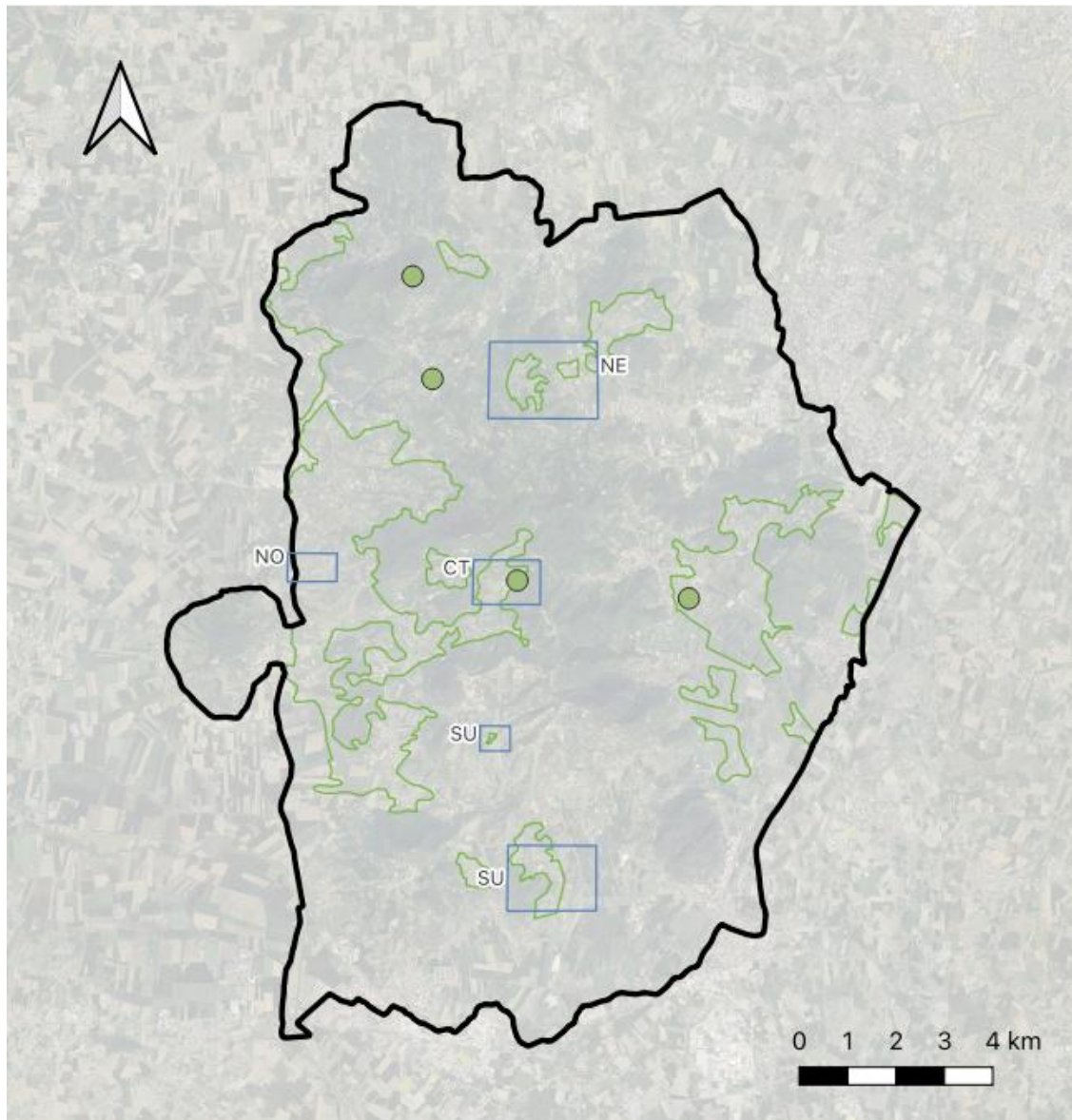
The figure consists of two maps. The left map shows the outline of Italy with a red dot in the northeast indicating the location of the study area. A dashed line connects this dot to a larger, more detailed map on the right. The right map shows the administrative regions of the Veneto area, labeled with two-letter codes: VI (Treviso), VR (Verona), PD (Padova), VE (Vicenza), and RO (Rovigo). A specific area within the PD region is highlighted with a blue and green color gradient, representing the study area.



- [illegible]

Image source: Parco Regionale dei Colli Euganei website

Meteorological Stations Location



Legend

- Areas
- Meteorological
- Euganean Hills Regional Park Delimitation
- Vineyard Land Cover (Corine 2018)
- Google Satellite

Not optimal for accurate analysis!

LANDSAT Collection for Climatology

Landsat Missions: Imaging the Earth Since 1972

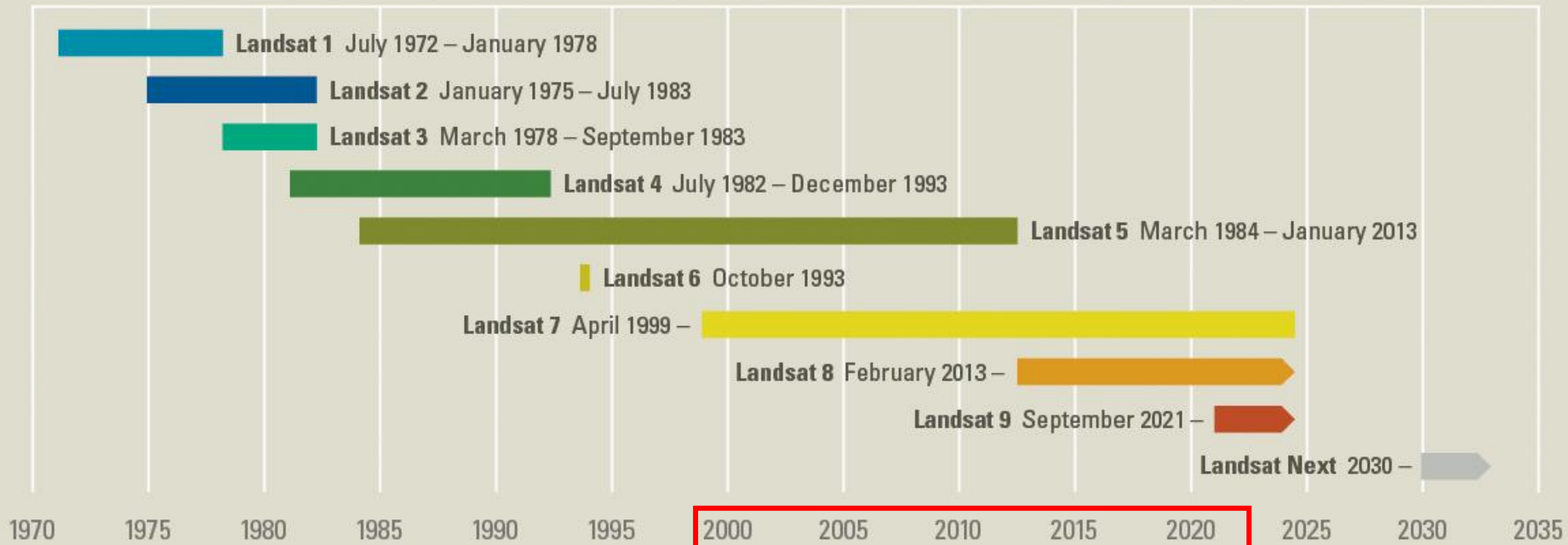


Image source: U.S. Geological Survey

Data Sources

➤ **LANDSAT Data:** Landsat 7, 8, 9 satellites were used to reconstruct a **pixel-based 20-year climatology** (2000 – 2023) using Multispectral and Thermal imagery

➤ **Agronomic Data:** Buds' fruitfulness data:

- Potential fruitfulness
- N. Blind buds
- N. Sterile buds
- **Varieties:** Cabernet sauvignon, Glera, Chardonnay, Merlot, Moscato Giallo.

➤ **Digital Terrain Model (DTM):** with a 5m resolution, from Veneto Regional Geoportal

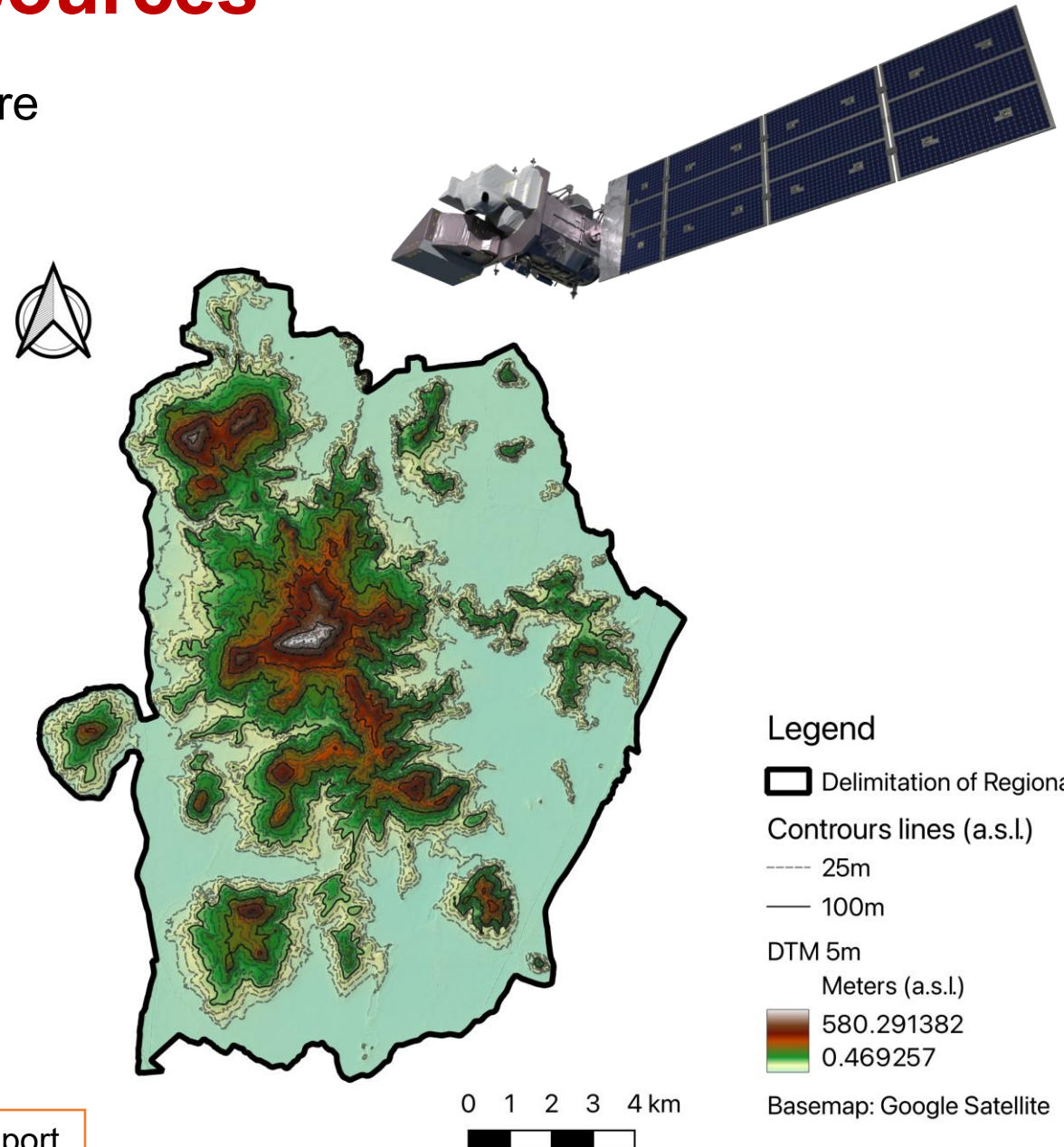
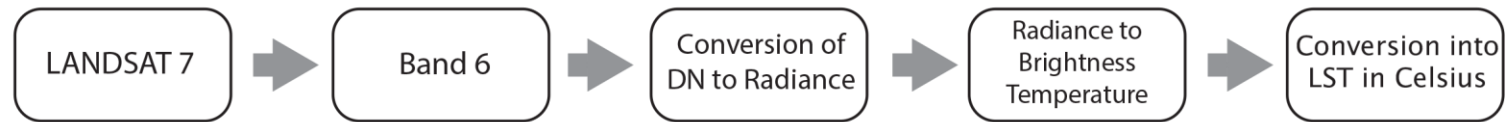


Image sources: U.S. Geological Survey and Final Report

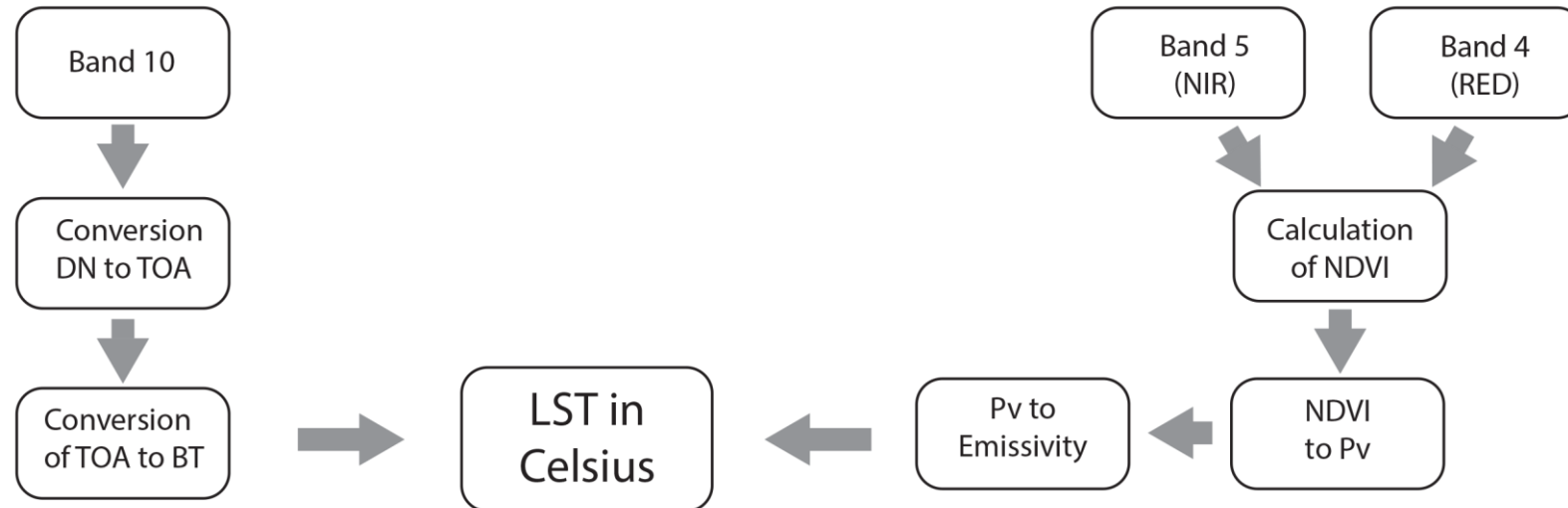
Methodology

➤ LST Derived from LANDSAT Satellites:

LANDSAT 7 procedure:



LANDSAT 8 and 9 procedure:



References:

- USGS Algorithm for Landsat 7,8 LST extraction is detailed in the Landsat 7,8 Science Data User's Handbook.

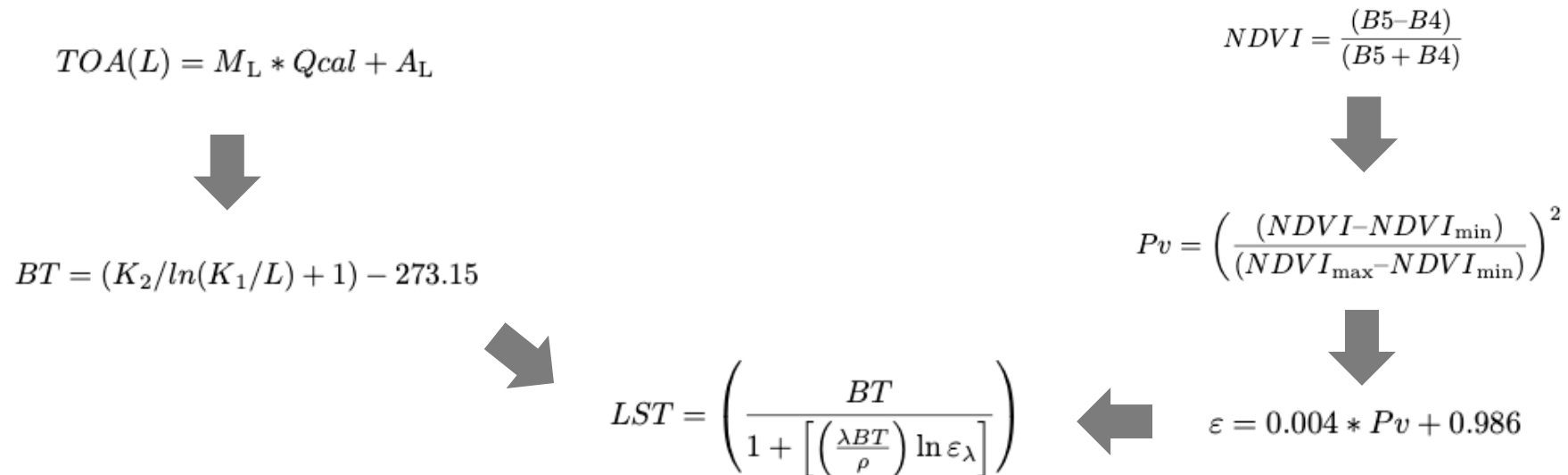
Methodology

➤ LST Derived from LANDSAT Satellites:

LANDSAT 7 procedure:

$$L_{\lambda} = \left(\frac{(L_{max_{\lambda}} - L_{min_{\lambda}})}{(Q_{cal_{max}} - Q_{cal_{min}})} \right) * (Q_{cal} - Q_{cal_{min}}) + L_{min_{\lambda}} \longrightarrow LST = (K_2 / \ln(K_1 / L) + 1) - 273.15$$

LANDSAT 8 and 9 procedure:



References:

- USGS Algorithm for Landsat 7,8 LST extraction is detailed in the Landsat 7,8 Science Data User's Handbook.

Methodology

Data Integration:

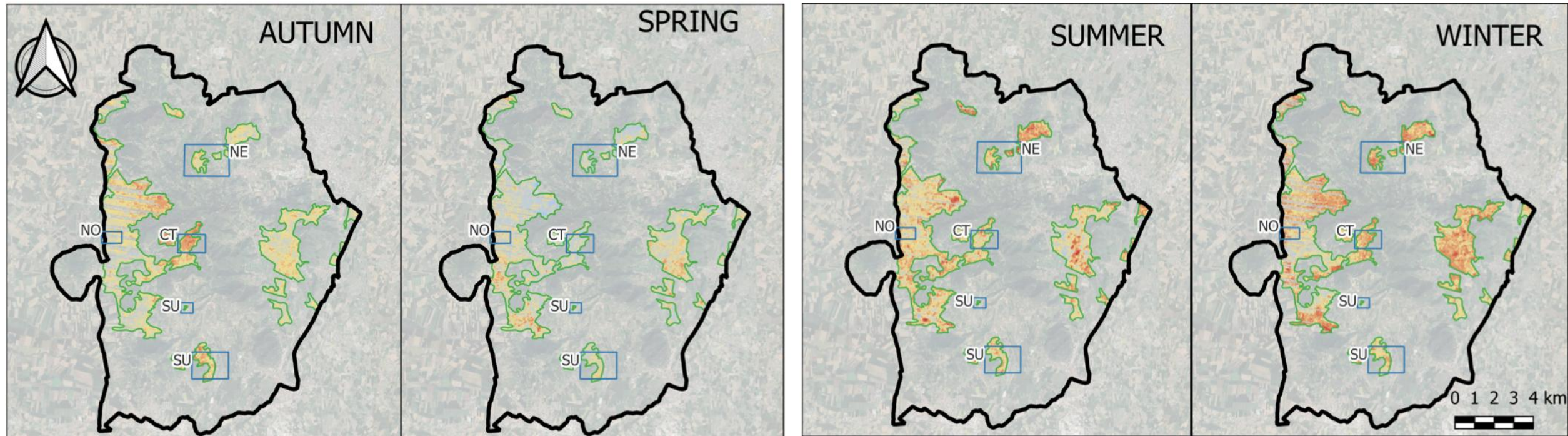
- I. **LST** and **DTM** derived **altitude** and **aspect** data were overlaid QGIS. LST was tested statistically against altitude and aspect using Python (*codes uploaded to a GitHub Repository*).
- II. **Temperature variability (*anomalies*)** across different areas was correlated **with bud fruitfulness, sterile buds, and blind buds' data**, using statistical analysis to identify significant trends (from 2018 to 2021)

Risk Mapping:

- I. LST data was used to develop **seasonal risk maps** (*reconstructed from a 20-year climatology*), highlighting areas at risk of temperature extremes (e.g., late frost or heat stress) that affect bud **fruitfulness**.

Results

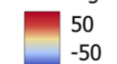
Risk Maps



Legend

Percentile

Signed Values Index



Corinne Land Cover: Vineyards

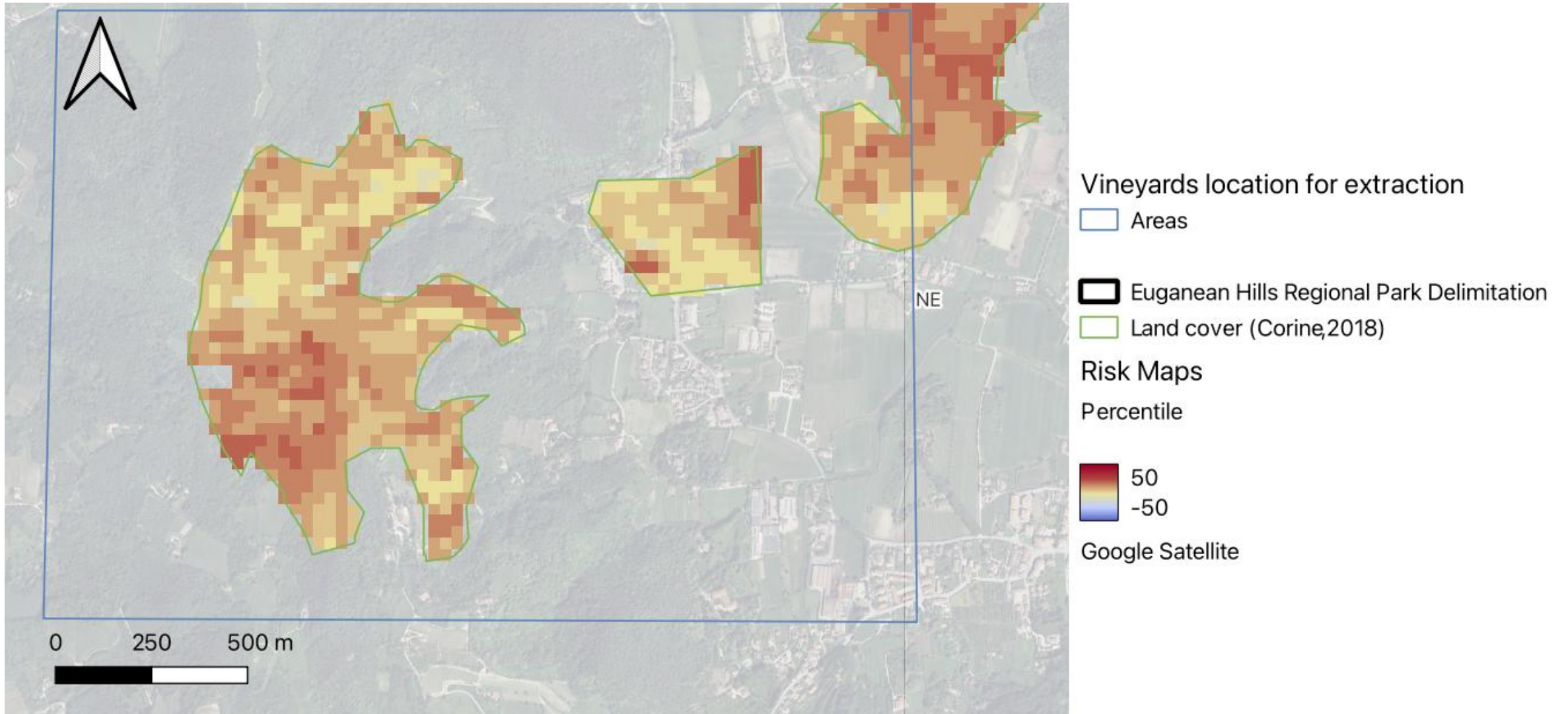
Delimitation of Regional Park

Areas

Base map: Google Satellite

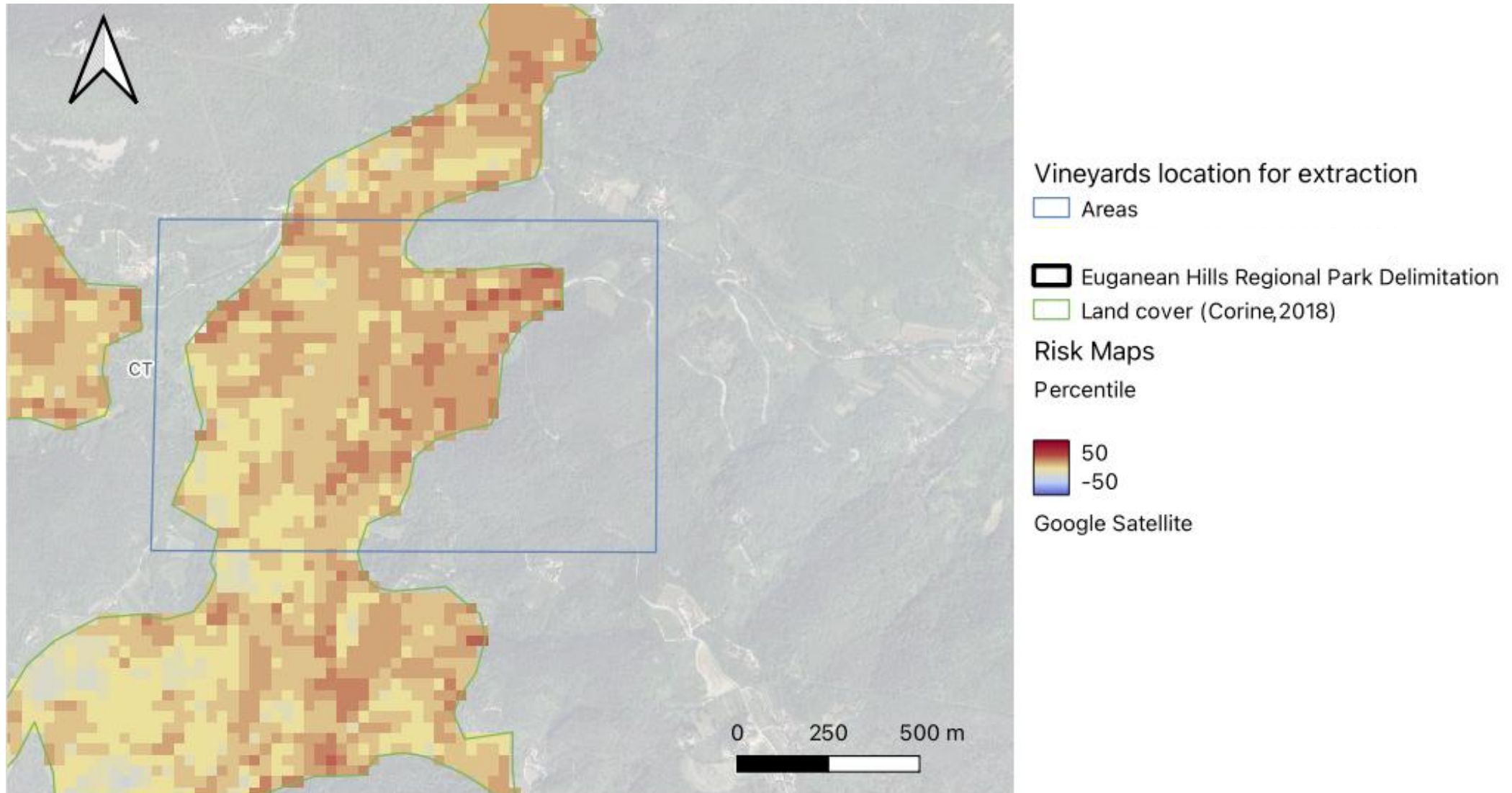
Results

Winter Map of North Area



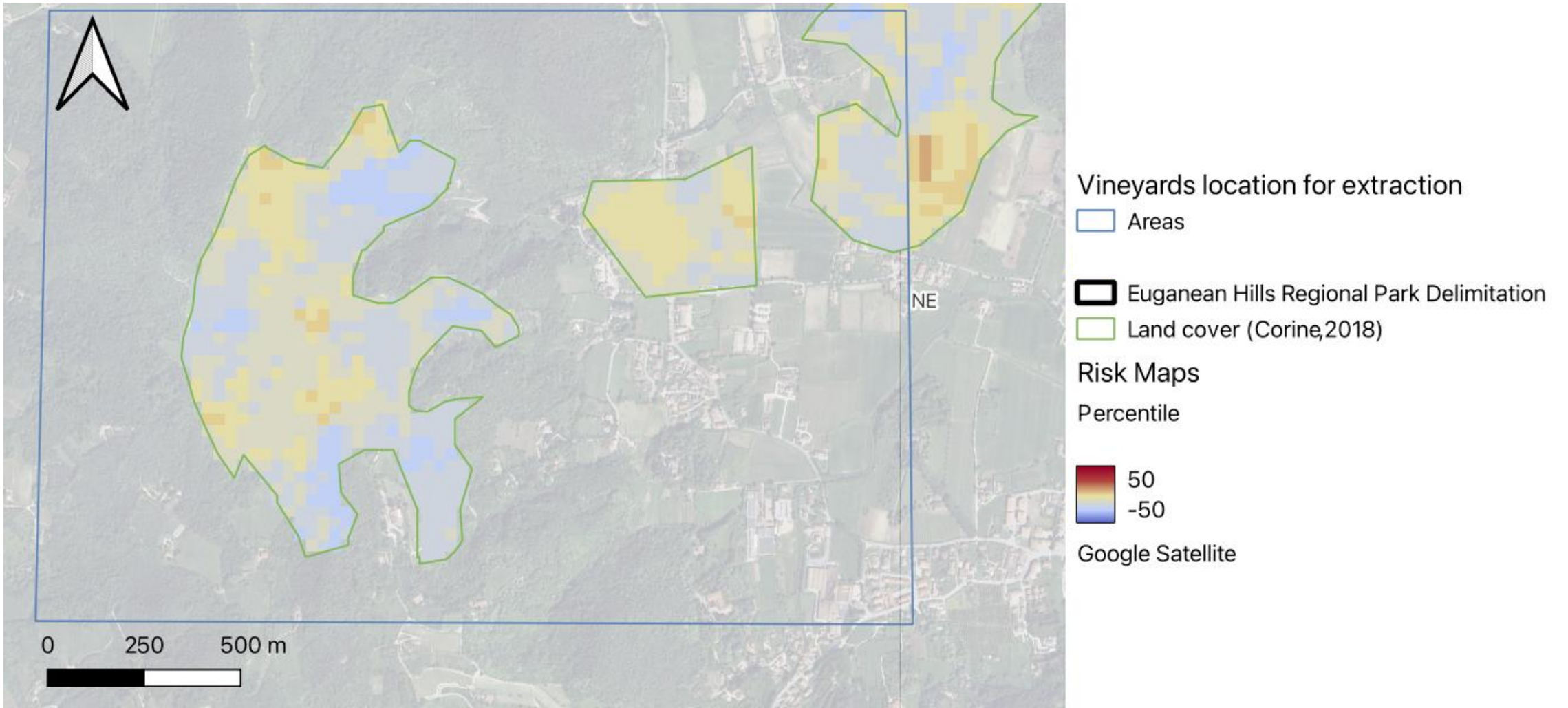
Results

Winter Map of Central Area



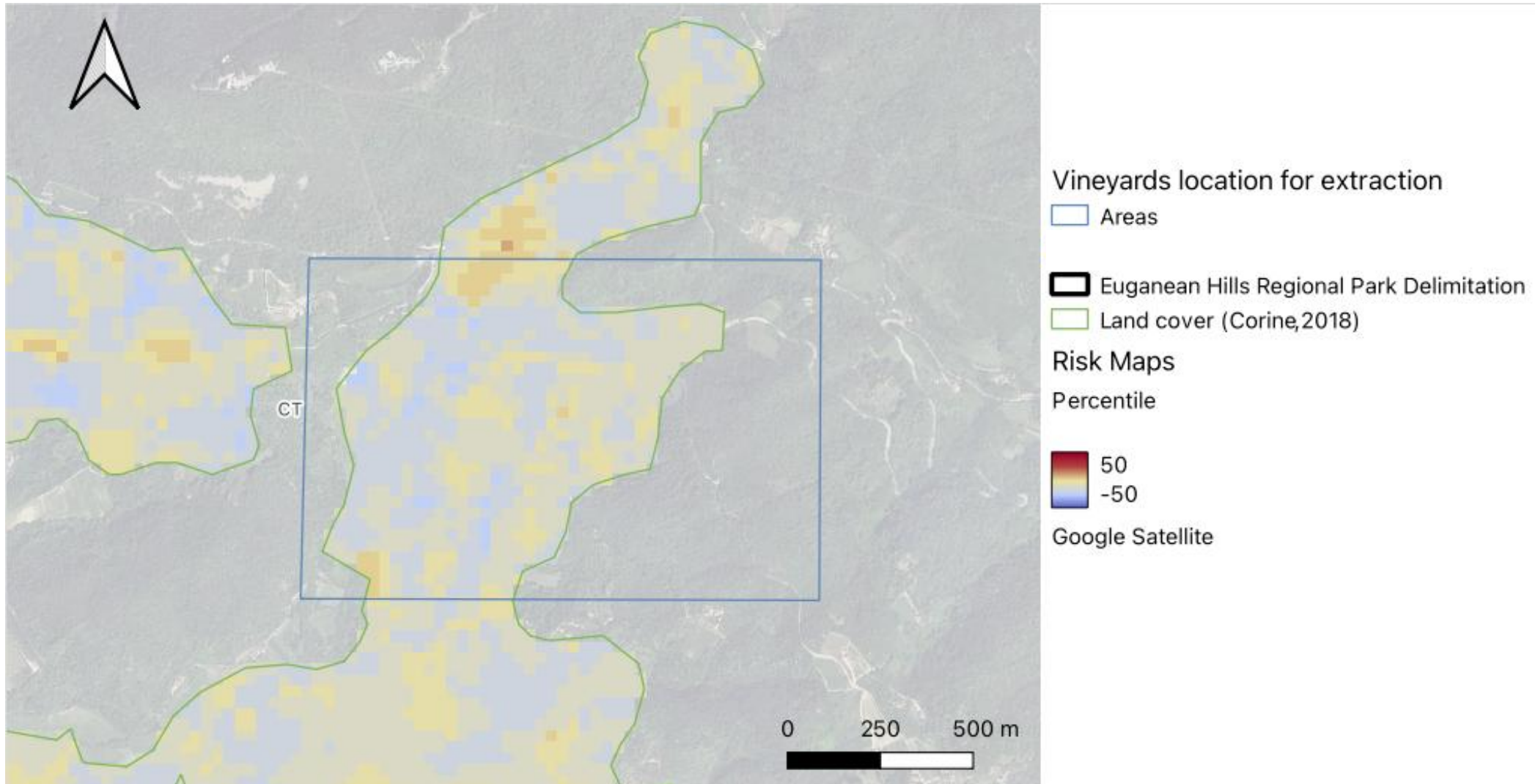
Results

Spring Map of North Area



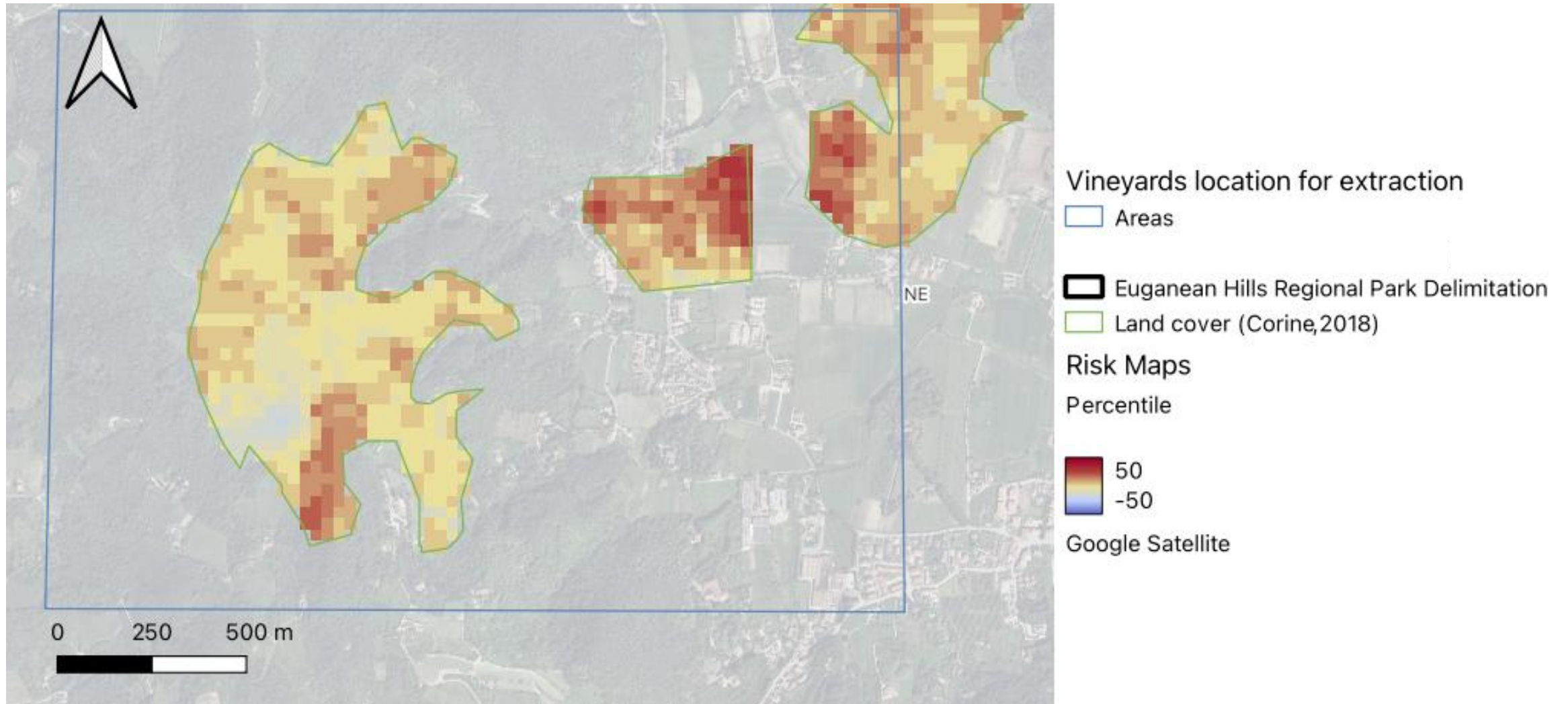
Results

Spring Map of Central Area



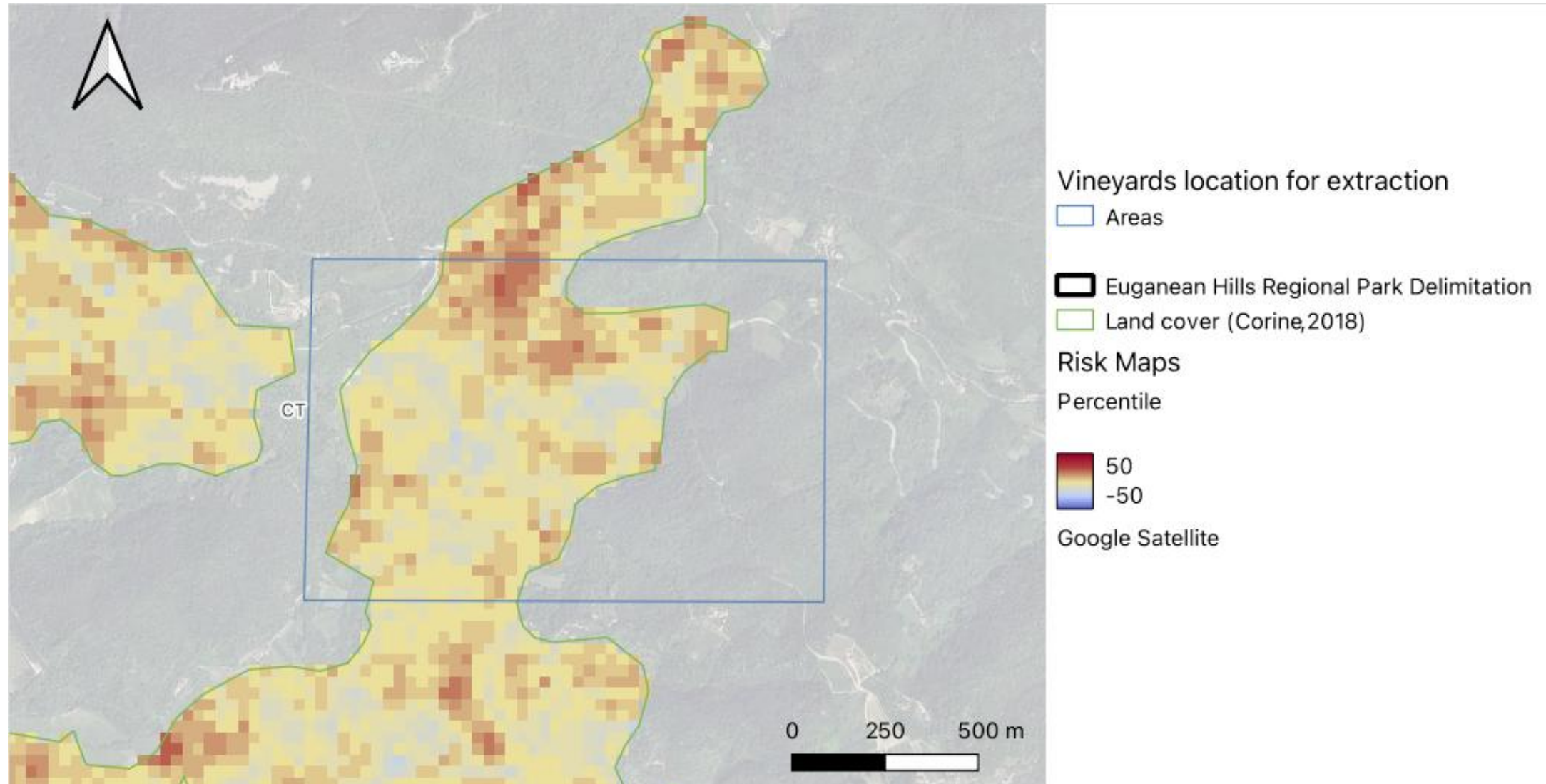
Results

Summer Map of North Area



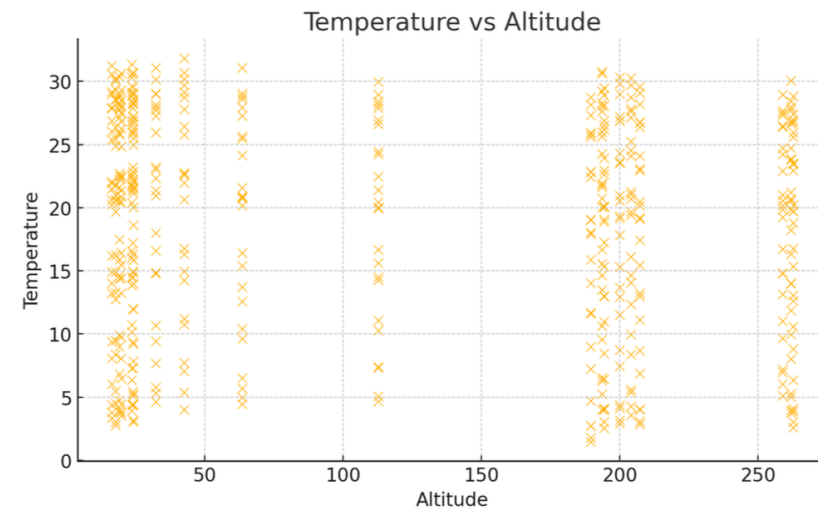
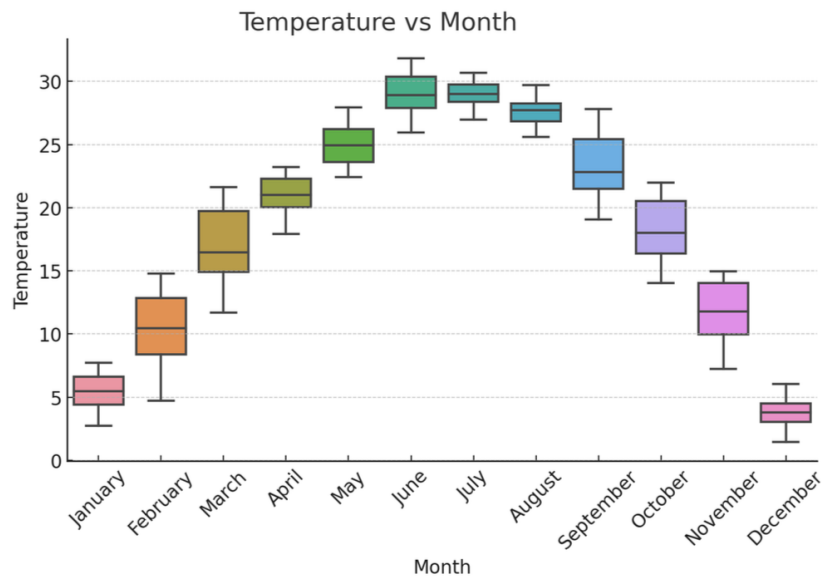
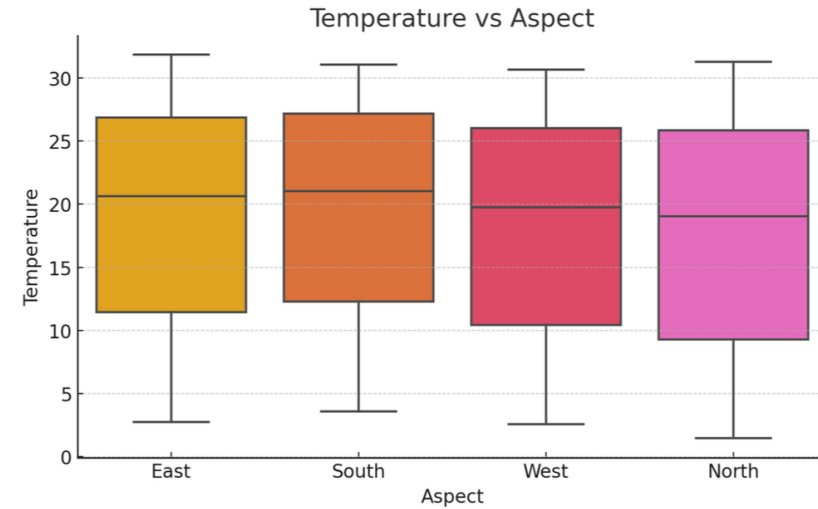
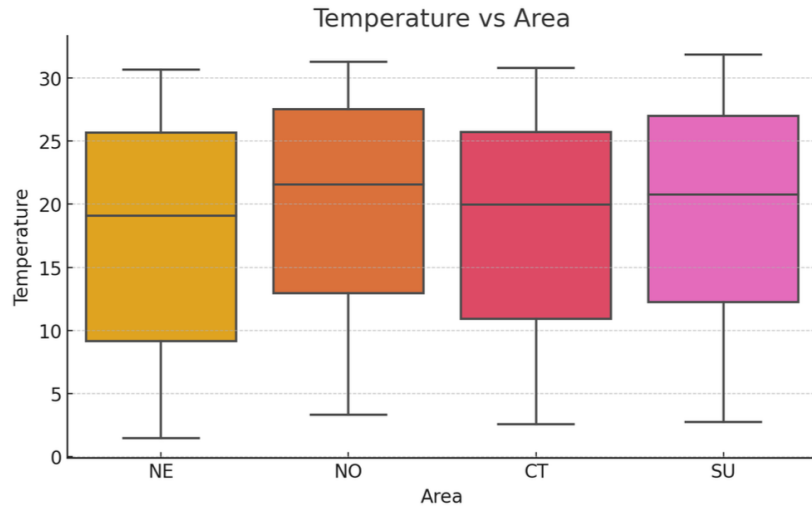
Results

Summer Map of Central Area



Results

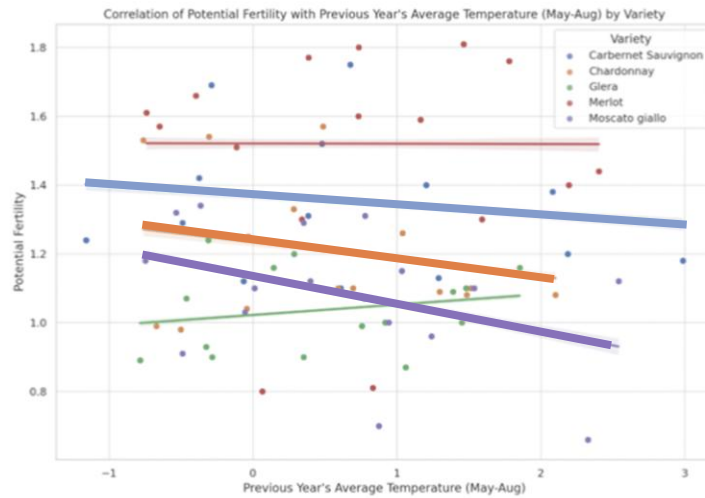
LST and DEM analysis: Climatology



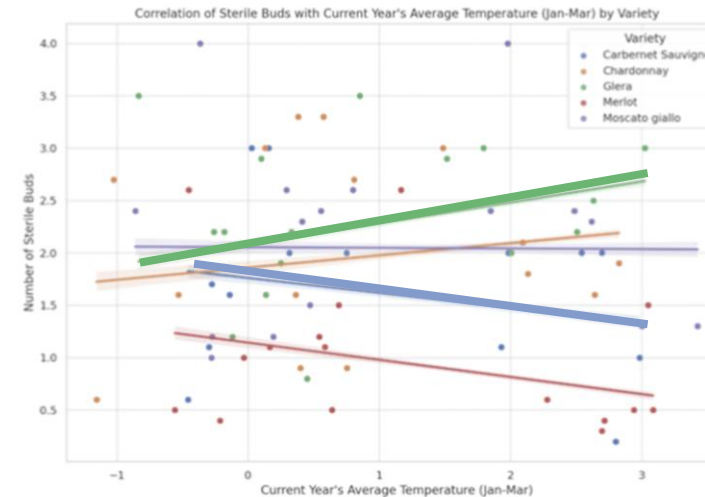
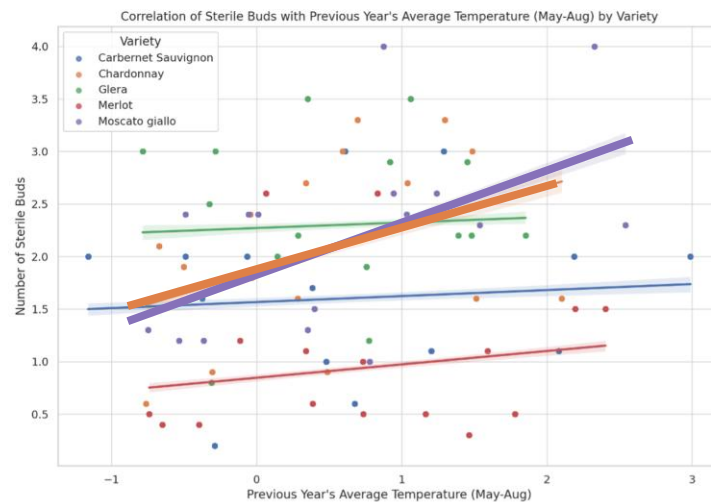
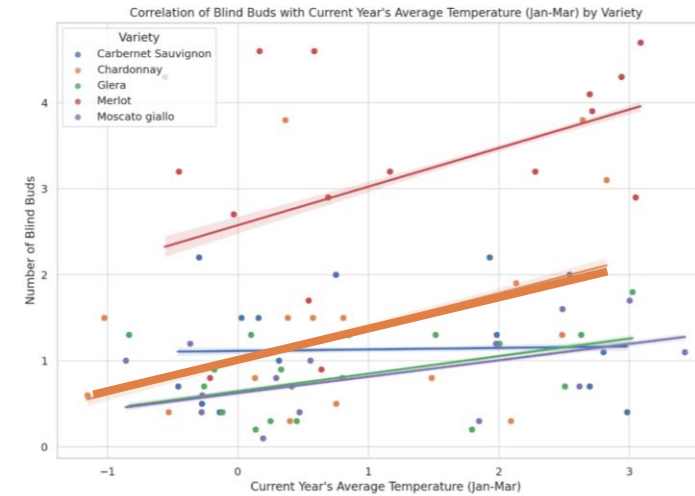
Results

Buds' Fruitfulness LST Correlations: 2018 - 2021

Previous year (May⁻¹-Aug⁻¹)



Current year (Jan-Mar)



Results

Temperature-Sensitive Varieties



Cabernet Sauvignon:

Sensitive to high-temperature anomalies. High temperatures increase sterile buds and decrease blind buds, affecting overall fertility negatively.



Chardonnay:

Sensitive to high-temperature anomalies, showing increased blind buds with rising temperatures (weak correlations).



Moscato Giallo:

Sensitive to both previous summer temperatures and winter temperatures. Warmer winters reduce blind buds but increase sterile buds, confirming its sensitivity to temperature fluctuations.



Glera:

Mixed Sensitivity: Sensitive to January temperatures (increases in blind buds) but more resilient to March temperature anomalies, where it performs better.

Results

Resilient Varieties:

Merlot:

Merlot shows minimal sensitivity to temperature anomalies, especially regarding sterile buds. It benefits from warmer March temperatures, reducing blind buds, confirming it as a resilient variety.



Conclusions

- I. **Variety-specific Responses:** Different grape varieties exhibit varying sensitivity to temperature anomalies.
- II. **Impact of Temperature Anomalies:** Temperature anomalies during critical growth periods strongly influence the occurrence of sterile and blind buds.
- III. **Risk Maps:** Using LST data, risk maps were created to identify regions within the vineyards at higher risk due to temperature fluctuations, allowing targeted management strategies.
- IV. **Adaptive Practices:** The study recommends adapting vineyard management.



Suggestion for Future Research

- **Integration of Additional Environmental Factors**
- **Use of Drone Technology**
- **Long-Term Data Collection with Drones**
- **Refinement of Predictive Models**
- **Climate Change Impact Analysis**
- **Multidisciplinary approach:**
 - I. Environmental Conservation and Ecosystem Monitoring**
 - II. Disaster Risk Reduction**
 - III. Climate Change Research and Policy**

Thank you for your attention!

enrico.magazzino@phd.unipd.it

